SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

C01_AT2 — DATA INTERPRETATION EXERCISE

Course Code:	DSA0606	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO1 –Apply (BL3)	**Assessment:**	Assessment Tool 2 – Data Interpretation
Weightage:	20% of CIA	**Total Marks:**	20(2 Questions × 20 Marks)
CO1 Statement:	*Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)*		
Student Name:	M.SAI HARSHITHA	**Reg. No.:**	192424408
Section / Batch:	B.TECH AI & DS	**Date:**	28.07.2026

Code of Conduct

I M.SAI HARSHITHA / 192424408(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.SAI HARSHITHA

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.

Annexure A – DATA INTERPRETATION

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Data Interpretation Exercise

CO1 · BT3: Apply ·

This rubric is used to assess student responses to the AT2 Data Interpretation Exercise problems, in which students interpret a described data visualization (data types, aesthetic mappings, scales, coordinate systems, and color scales) and answer accompanying sub-questions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough, accurate understanding of data types (nominal, ordinal, numerical), aesthetic mappings, scales, coordinate systems, and color scales as they apply to the given visualization scenario.	Good understanding of the relevant visualization concepts, with only minor gaps in identifying data types, aesthetic mappings, or scale choices.	Basic understanding of visualization concepts, but with some misconceptions about data types, aesthetics, coordinate systems, or color scales.	Poor understanding of core visualization concepts; major misconceptions about data types, aesthetics, coordinate systems, or color scales.
Explanation	**25%**	Provides detailed, well-reasoned explanations for each sub-question, using specific details from the scenario and correct visualization terminology.	Provides clear explanations with adequate justification, referencing the scenario appropriately for most sub-questions.	Explanations are limited or superficial; lacks depth or fails to consistently connect reasoning to the given scenario.	Explanations are poor, unclear, or missing; answers are unsupported assertions with no reasoning shown.
Application	**20%**	Effectively applies visualization concepts to interpret the specific scenario, drawing accurate, well-reasoned conclusions from the described	Applies concepts to interpret the scenario correctly, with only minor errors in reasoning or in the conclusions drawn.	Limited application of concepts to the scenario; conclusions are weakly supported or only partially correct.	Fails to apply concepts to the scenario, or the conclusions drawn are incorrect or irrelevant to the data shown.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		chart or data pattern.			
Organization	**15%**	Answers are well-structured, addressing each sub-question (a, b, c) in a clear, logical sequence with coherent flow throughout.	Answers are organized and address each sub-question, with only minor issues in flow or clarity.	Basic structure; sub-questions are addressed but answers lack clarity or a logical sequence.	Poorly organized; answers are incoherent, and sub-questions are left unaddressed or answered out of order.
Accuracy	**10%**	All parts of the answer — data type identification, aesthetic/scale justification, and final interpretation — are completely accurate.	Mostly accurate, with only minor omissions or a small error in one part of the answer.	Partially correct; some key points (e.g., data type, aesthetic justification, or interpretation) are missing or incorrect.	Answers are largely incorrect, with major gaps in identifying data types, aesthetics, scales, or drawing conclusions.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

AT2 — Data Interpretation Exercise

NAME: M.SAI HARSHITHA
REG.NO:192424408
COURSE CODE: DSA0606
COURSE NAME:DATA HANDLING AND VISUALIZATION

Problem 6 (10 Marks)

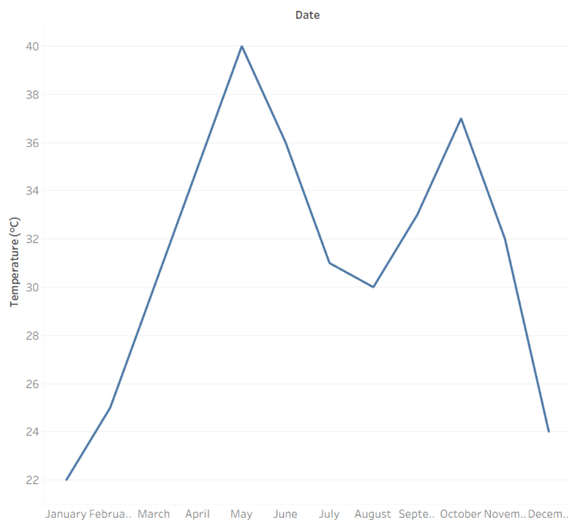
Scenario: A weather dataset includes Temperature (numeric, °C), recorded daily for a year, visualized as a line chart with Date on the x-axis and Temperature on the y-axis.

a) What aesthetic is used to map "Date," and what type of data is it?

b) Would a scale mapping "Temperature" to bar height be equally effective as mapping it to line position? Justify.

c) If the line shows two peaks in a year, what season-related insight can you extract from this pattern?

Date	Temperature (°C)
01-Jan-2025	22
01-Feb-2025	25
01-Mar-2025	30
01-Apr-2025	35
01-May-2025	40
01-Jun-2025	36
01-Jul-2025	31
01-Aug-2025	30
01-Sep-2025	33
01-Oct-2025	37
01-Nov-2025	32
01-Dec-2025	24



Daily Temperature Over One Year.

a) What aesthetic is used to map "Date," and what type of data is it?

Item	Answer
Chart Type	Line Chart
Aesthetic Used	Position on the X-axis
Data Type	Temporal (Date/Time)
Reason	The Date field is placed on the X-axis to show temperature changes over time.

b) Would a scale mapping "Temperature" to bar height be equally effective as mapping it to line position? Justify.

Answer

No. A line chart is more effective because temperature changes continuously over time. The line clearly shows trends, increases, decreases, and seasonal variations. A bar chart is more suitable for comparing separate categories than continuous time-series data.

c) If the line shows two peaks in a year, what season-related insight can you extract from this pattern?

Answer

Two peaks indicate that the temperature reaches high values during two different periods of the year. This suggests the region experiences two warm seasons or two periods of maximum temperature due to seasonal climatic variations.

Problem 16 (10 Marks)

Scenario: A choropleth map shows population density by country using a sequential color scale from light yellow (low density) to dark green (high density).

a) Is a sequential scale an appropriate choice here? Justify your answer using the nature of the data.

b) If Country A appears very dark green and Country B appears very light yellow, what can you conclude about their relative population densities?

c) What potential misinterpretation could occur if a viewer is colorblind and cannot easily distinguish yellow from green tones?

a) Is a sequential scale an appropriate choice?

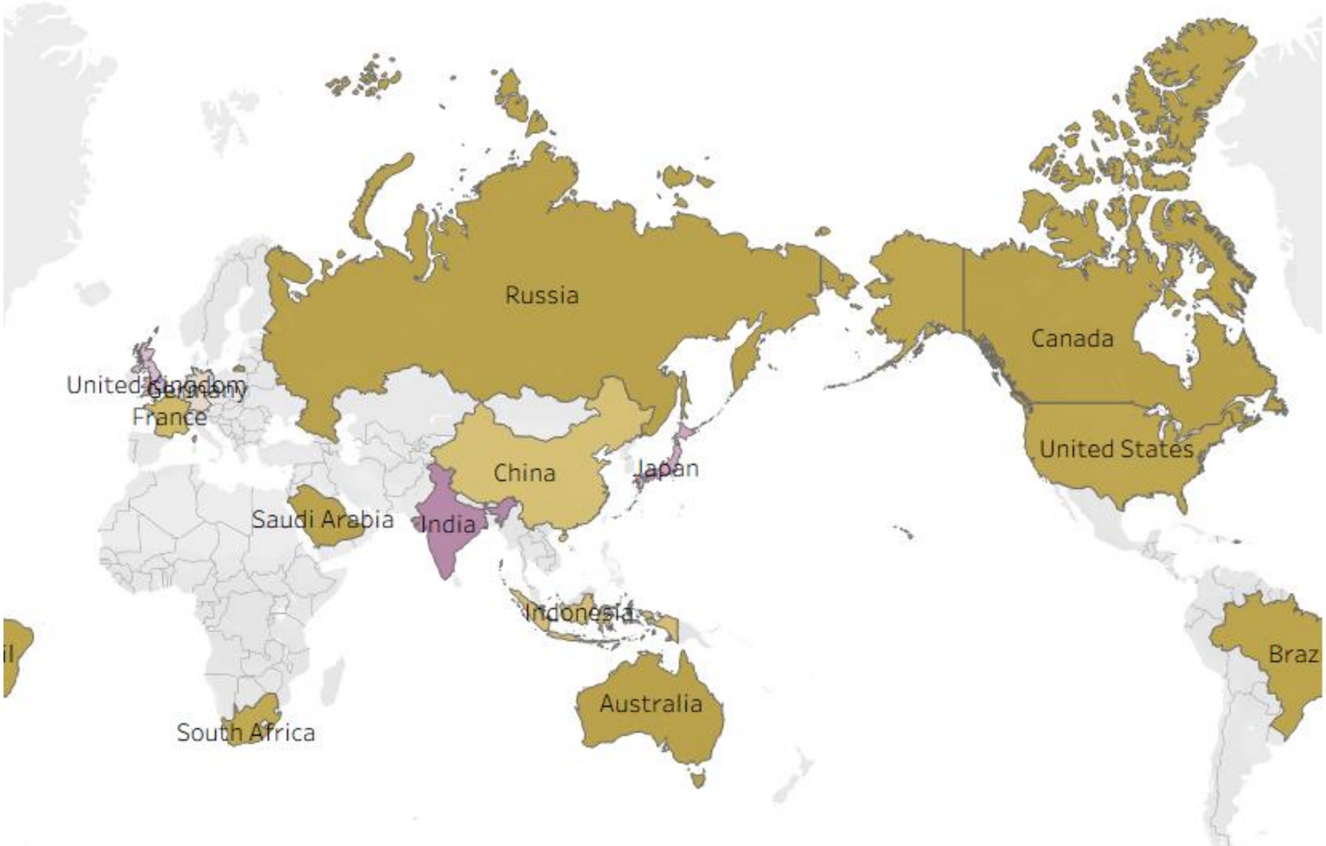
A sequential color scale is appropriate because population density is continuous numerical data ranging from low to high values. Light yellow represents low population density, while dark green represents high population density, making differences easy to understand.

b) Country A vs Country B

If Country A appears dark green and Country B appears light yellow, it indicates that Country A has a much higher population density than Country B.

c) Colorblind Interpretation

A viewer with color vision deficiency may have difficulty distinguishing between yellow and green shades. This could lead to incorrect interpretation of population density. To reduce this problem, use colorblind-friendly palettes, labels, or tooltips along with the color scale.



POPULATION ACROSS COUNTRIES